

debugging. Embedded design is the buzz word today, informs Appalla. He says, "Former truly-isolated functional blocks and circuit components now get closely integrated on circuit board or even multi-chip level. For system debugging, a time-correlated analysis of analogue signals, such as switched mode power supplies or D/A converters, as well as digital signals of various communication and programming interfaces is required. Additional complexity is given by protocol-based interfaces or integrated RF components."

The user requirements, as mentioned above, are drivers for standard oscilloscopes. "Debugging of protocol-based interfaces, such as I2C, SPI or CAN, is supported with trigger and decoded options. Often a bandwidth of 1GHz is required, not only because of faster digital signals but also driven by fast edges of clock signals or power supply switching. A suitable sample rate of 5Gsample/s and a minimum acquisition memory of 10Msamples enable a signal acquisition with high resolution over a long observation period," adds Appalla.

In general, the trends in oscilloscopes are focused on helping engineers answer their questions faster. In addition to integrating logic analyser and spectrum analyser functions in the MSO and MDO classes of instruments, another trend is the integration of signal sources in the oscilloscope, informs Aparo. She says, "This function can range from a simple sine-wave or square-wave generator to something as complex as a multichannel arbitrary waveform generator that allows users to create modulated signals. This trend of integrating more features and functions into the oscilloscope helps engineers get answers more quickly, because it provides all the needed tools in a single platform." An example of this kind of integration can be witnessed in Tektronix's MDO that boasts of six instruments built in one, namely, oscilloscope, spectrum analyser, logic analyser, arbitrary/function generator, protocol analyser and a digital voltmeter.

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Oscilloscopes also increasingly offer better bandwidth, more analysis capabilities and better hardware specifications, informs Bhatia. He says, "Analysis capabilities offered today include variety of protocol analysis tools, compliance applications and more mathematical functions. Tools such as AC calibration of probes, de-embedding and equalisation offer better reproduction of the signals on the scopes."

New and still not so common with oscilloscopes is the operation in the frequency domain. But users request this functionality, informs Appalla. He says, "They want, for example, to debug EMI problems caused by power supplies, fast edges or touch-screen emission."

## Technologies for tomorrow

One of the new technologies developed for high-bandwidth scopes is indium phosphide (InP) semiconductors, informs Bhatia. He says, "New technologies are developed by Agilent labs (a research organisation within Agilent), which focuses on higher integration, low-noise front ends and high-speed designs." He adds, "InP semiconductors have very low noise and high transistor-breakdown voltage at high frequencies. Thus, they can handle higher voltages at high bandwidths. This helps design engineers make high-bandwidth measurements very accurately and with great confidence."

Targeting embedded design engineers, Tektronix delivered advanced 802.11 WLAN test solutions to its mixed-domain and performance oscilloscopes. Engineers can use tools for inte-

grating 802.11 a/b/g/j/n/p/ac WLAN connectivity into their product designs. The new WLAN solutions address the growing demand toward integrating Wi-Fi in everything from home appliances to industrial equipment.

Many of the new technologies being developed are also software improvements that make it easier for engineers to test or simulate their systems more quickly. Aparo says, "This is often found in larger markets like wireless communications or automotive radar, where there is a large pool of engineers who are working on these applications around the world and their needs converge on common requirements or published standards." Talking about how it would help a design engineer, she adds, "Improving software for oscilloscopes can reduce the time it takes a design engineer to understand what is not working or verify the performance of a new platform. By pre-loading common tests or waveforms into the software, engineers know there is consistency in the information they are gathering, and this makes it easier to collaborate with other team members."

High-end test and measurement requires high-end semiconductors to make them work efficiently. Mantri says, "So, support on making the semiconductors application-specific from respective vendors can create a big wave in this market. Compact USB digital oscilloscope will be a dream come into reality for service engineers on field at an affordable cost." ●

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